INCH-POUND

MIL-M-28787/123A(SH)
AMENDMENT 2
24 August 1990
SUPERSEDING
AMENDMENT 1
28 December 1982

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FLIP-FLOP, JK, DIGITAL, KEY CODE ADH

This amendment forms a part of Military Specification MIL-M-28787/123A(SH), dated 22 June 1982, and is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 4

TABLE I. Delete and substitute new table I as printed on page 3 of this amendment.

PAGE 8

* 4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through MM shall be performed in any order with the following exceptions. Test conditions A, B, C, and D are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, and D where applicable, followed by the remaining tests in the sequence."

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* 4.5.1, fifth and sixth lines: Delete "E, F1, G through KK, and MM" and substitute "E through Y, AA through FF, and JJ".

* 4.5.3, third and fourth lines: Delete "E, F1, G through KK, and MM" and substitute "E through Y, AA through FF, and JJ".

* 4.5.3, sixth line: Delete "E, F1 and G through MM" and substitute "E through Y, AA through FF, and JJ".

PAGE 11

* FIGURE 2, NOTE 1: Delete "R1" and substitute "R11".

* FIGURE 2, NOTE 2: Delete "R2-R11" and substitute "R1-R10".

PAGE 23

TABLE V: Delete and substitute new table V as printed on page 4 of this amendment.

PAGE 63

TABLE XIX: Delete and substitute new table XIX as printed on page 5 of this amendment.

PAGE 64

FIGURE 27, ITEM D05: Add "E" next to the arrow on the line from CHAN A of ITEM D05.

FIGURE 27, NOTE 5, second line: Delete "item V03" and substitute "item M02".

PAGE 71

* FIGURE 32: Delete test point "A" and substitute "C"

* FIGURE 32: Delete test point "B" and substitute "A".

* FIGURE 32: Delete test point "C" and substitute "D".

* FIGURE 32. Delete test point "D" and substitute "B".

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25 ± 5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A25	Zener applied potential, cond A	III	S	3.8	5.5	3.5	5.5	3.8	5.5	3.5	5.5	V
B1-B5	Zener applied potential, cond B	III	S	3.8	5.5	3.5	5.5	3.8	5.5	3.5	5.5	V
C1-C10	V_{OUT} (low-loaded)	IV	S	0.0	1.1	0.0	1.1	0.0	1.1	0.0	1.1	V
D	Isolation	V	S	10	∞	1	∞	10	∞	1	∞	MΩ
E	Resistance	V	X	4	5	4	5	4	5	4	5	kΩ
F1	Continuity	V	X	0	1	0	1	0	1	0	1	Ω
F2	Continuity	V	S	0	1	0	1	0	1	0	1	Ω
G1-G5	Input current, reverse clock	VI	X	0	4	0	8	0	6	0	10	µA
H1-H10	Input reverse current, synchronous	VII	X	0	2	0	4	0	3	0	5	µA
J1-J10	Input reverse current, asynchronous	VII	X	0	4	0	8	0	6	0	10	µA
K1-K10	Input forward current	VIII	X	0.20	1.07	0.20	1.07	0.20	1.07	0.20	1.07	mA
L1-L10	Input forward current	VIII	X	0.2	3.2	0.2	3.2	0.2	3.2	0.2	3.2	mA
M1-M10	Input forward current	VIII	X	0.2	3.2	0.2	3.2	0.2	3.2	0.2	3.2	mA

See footnotes at end of table.

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TABLE V. Isolation, resistance, and continuity test.

Test condition	Parameter	Connect				Initial limits 25 ± 5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
D	Isolation	11	All other pins	Open	Open	10	∞	MΩ	9
E	Resistance	Open	Open	1	32	4	5	kΩ	9
F1	Continuity	Open	Open	10	30	0	1	Ω	9
F2	Continuity	Open	Open	11	Frame	0	1	Ω	9

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TABLE XIX. Transient immunity (J and K) and minimum pulse sensitivity (J-K) test.

Test cond	Parameter	Connect					Initial limits 25 ± 5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
GG1	1K Transient immunity, J and K	7	4	3	2	Open	0.00	0.40	V	27, 28
GG2	2K Transient immunity, J and K	19	16	15	14	Open	0.00	0.40	V	27, 28
GG3	3K Transient immunity, J and K	22	25	26	27	Open	0.00	0.40	V	27, 28
GG4	4K Transient immunity, J and K	29	9	28	12	Open	0.00	0.40	V	27, 28
GG5	5K Transient immunity, J and K	35	38	39	40	Open	0.00	0.40	V	27, 28
GG6	1J Transient immunity, J and K	7	8	5	6	Open	0.00	0.40	V	27, 28
GG7	2J Transient immunity, J and K	19	20	17	18	Open	0.00	0.40	V	27, 28
GG8	3J Transient immunity, J and K	22	21	24	23	Open	0.00	0.40	V	27, 28
GG9	4J Transient immunity, J and K	29	31	13	33	Open	0.00	0.40	V	27, 28
GG10	5J Transient immunity, J and K	35	34	37	36	Open	0.00	0.40	V	27, 28
HH1	1J-1K Minimum pulse sensitivity, J-K	7	8,4	Open	Open	3	1.6	2.4	µS	27, 29
HH2	2J-2K Minimum pulse sensitivity, J-K	19	20,16	Open	Open	15	1.6	2.4	µS	27, 29
HH3	3J-3K Minimum pulse sensitivity, J-K	22	21,25	Open	Open	26	1.6	2.4	µS	27, 29
HH4	4J-4K Minimum pulse sensitivity, J-K	29	31,9	Open	Open	28	1.6	2.4	µS	27, 29
HH5	5J-5K Minimum pulse sensitivity, J-K	35	34,38	Open	Open	39	1.6	2.4	µS	27, 29

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